

CSA1513 · CLOUD COMPUTING AND BIG DATA ANALYSIS

Identity and Access Management (IAM) in Cloud Environments

Authentication, Authorization Architectures, Security Benefits, and Enterprise Implementations

Pratip Raj.P (192521409)

Faculty: Antony Joseph Rajan

Course: CSA1513 – Cloud Computing and Big Data Analysis

Agenda

What we'll cover in this session

01

Introduction to IAM

Why identity is the new security perimeter for the cloud

02

Authentication Mechanisms

MFA, SSO, federated identity, biometrics

03

Authorization Architectures

RBAC, ABAC, PBAC and Zero Trust models

04

IAM Architecture

How the components interact end to end

05

Security Benefits

Risk reduction, compliance, visibility

06

Enterprise Implementation

AWS, Azure, GCP real-world practice

Introduction to IAM

Identity and Access Management is the security discipline that ensures the right individuals and services access the right resources, at the right time, for the right reasons — in dynamic, distributed cloud environments.



Identity as the perimeter

In the cloud, the traditional network boundary disappears — identity becomes the primary control point.



Two core pillars

Authentication (who are you) and Authorization (what can you do) work together in every access decision.



Applies to humans & machines

IAM governs employees, admins, third parties, APIs, and workloads alike.

81%

of cloud data breaches involve compromised or misused credentials

Why it matters now

Multi-cloud sprawl, remote access, and API-driven workloads mean identities now outnumber servers many times over — making disciplined IAM essential to enterprise cloud security.

Authentication

Verifying “Who are you?” before granting any access



MFA

Multi-Factor Authentication

Combines password, device token, and biometrics to block credential-only attacks.



SSO

Single Sign-On

One authentication event grants access across multiple trusted cloud applications.



FED

Federated Identity

SAML / OAuth2 / OpenID Connect let users authenticate once across organizational boundaries.



BIO

Biometric & Adaptive Auth

Fingerprint, face ID, and risk-based signals (location, device) adjust verification strength.

Authorization Architectures

Deciding “What can you do?” once identity is confirmed

1

RBAC

Role-Based Access Control

Permissions are tied to job roles (Admin, Developer, Viewer) — simple to manage at scale.

2

ABAC

Attribute-Based Access Control

Access decisions use attributes: department, resource tag, time of day, device posture.

3

PBAC

Policy-Based Access Control

Centralized, machine-readable policies (JSON/XML) evaluated dynamically per request.

4

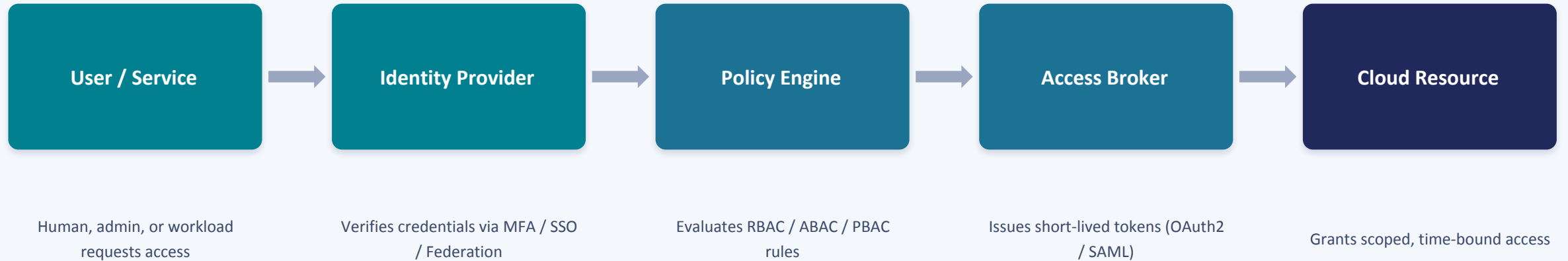
Zero Trust

Never Trust, Always Verify

No implicit trust by network location — every request is continuously authenticated and authorized.

IAM Architecture Diagram

End-to-end flow from user request to resource access



Continuous Monitoring, Logging & Audit — every access event is recorded and fed back to the Policy Engine for anomaly detection

Directory Services (e.g., LDAP / Active Directory) and centralized logging underpin every stage of this flow.

Security Benefits of IAM



Reduced Attack Surface

Least-privilege access limits what any single compromised identity can reach.



Regulatory Compliance

Supports GDPR, HIPAA, SOC 2 and ISO 27001 through auditable access controls.



Centralized Visibility

Unified dashboards track who accessed what, when, and from where.



Faster Incident Response

Instant credential revocation and session termination during a breach.



Improved User Experience

SSO and adaptive auth reduce login friction without weakening security.



Scalable Governance

Automated provisioning/deprovisioning keeps access aligned as teams grow.

Enterprise Implementations

How leading cloud platforms put IAM into practice

AWS IAM

- Users, Groups, Roles & Policies (JSON)
- Fine-grained resource-level permissions
- Integrates with AWS Organizations & SSO

Azure Active Directory

- Conditional Access policies
- Seamless hybrid on-prem/cloud identity
- Built-in MFA and Privileged Identity Mgmt

Google Cloud IAM

- Predefined, custom & basic roles
- Resource hierarchy-based policy inheritance
- Context-Aware Access for Zero Trust

Challenges & Best Practices

Common Challenges

- ✗ Privilege creep from unreviewed role assignments
- ✗ Fragmented identity across multi-cloud platforms
- ✗ Shadow IT and unmanaged API credentials
- ✗ Balancing strong security with user convenience

Best Practices

- ✓ Enforce least-privilege and periodic access reviews
- ✓ Adopt Zero Trust with continuous verification
- ✓ Centralize identity via a single trusted directory
- ✓ Automate provisioning, deprovisioning & rotation

Conclusion

IAM is the backbone of cloud security — binding authentication and authorization into a single, continuously enforced trust framework.

1

Strong authentication (MFA, SSO, federation) confirms identity before any access is granted.

2

Modern authorization models — RBAC, ABAC, PBAC, Zero Trust — enforce least-privilege at scale.

3

Enterprise platforms (AWS, Azure, GCP) operationalize IAM through policies, roles, and conditional access.

4

Well-governed IAM reduces breach risk, ensures compliance, and enables secure, scalable cloud adoption.

Thank You

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